# Lab 2: Jira Step-by-Step Guide

Clinical Trial Subject Recruitment Engine | PES1UG25CS845

## Read this first

**What is already done:** the whole backlog is written for you (5 Epics, 17 User Stories, priorities, Fibonacci story points, 2 sprint plans, the progress schedule and the reflection answers). Everything is in this guide and in SE_Lab2_Jira_Report.docx.

**What is NOT done yet:** nothing has been created inside your Jira account. I cannot log into Jira, so you must do the clicking. The screenshots and the real Jira burndown charts can only come from your own Jira, so the report has empty boxes for them.

**Time needed:** about 60 to 90 minutes if you copy and paste the text from the tables below.

### Words you need to know

- **Epic:** a big theme of work (for example "Account Registration & Access").
- **User Story:** one small feature written as "As a..., I want..., So that...". Each story belongs to one Epic.
- **Story Points:** a number (1, 2, 3, 5, 8 or 13) showing how big a story is. Bigger number = more work or more uncertainty.
- **Sprint:** a fixed time period (here 1 week) in which you finish a chosen set of stories.
- **Backlog:** the list of all stories not yet finished. The Backlog tab is where you create everything.
- **Burndown chart:** a graph of story points still remaining in a sprint. It should fall to 0 by the end of the sprint.

**Note:** Jira changes its screens from time to time. If a button is in a slightly different place from what is written here, look for the same name nearby. The screens in the lab handout are the reference.

## Step 0: Get ready

1. Open a browser (Chrome is best) and sign in to your Jira account. If you have none, sign up for free with your email at atlassian.com/software/jira.
2. Make a folder on your computer called Lab2_Screenshots. You will save 8 screenshots there.
3. Learn the screenshot shortcut. Windows: press Windows + Shift + S, then drag over the area. Mac: press Command + Shift + 4, then drag over the area.

## Step 1: Create the Scrum project (handout Step 1)

1. After logging in, look at the left sidebar. Click the + (plus) icon next to "Projects", then click "Create project".
2. In Project templates, click "Software development" on the left, then choose "Scrum" and click "Use template".
3. Choose the Company-managed project type (click "Select a company-managed project"). If your account only offers Team-managed, choose that and tell your instructor; the Scrum board works the same way.
4. On the "Add project details" page type: **Name** = Clinical Trial Subject Recruitment Engine and **Key** = CTRE. Click Next.
5. Jira shows your empty Scrum board. Click the "Backlog" tab (top menu). This is where you will do Steps 2 to 6.

## Step 2: Create the 5 Epics (handout Step 2)

1. On the Backlog tab, click the blue "Create" button at the top (next to the search bar).
2. In "Work type", choose Epic.
3. Copy the Summary and Description for Epic 1 from the table below. Paste them in. Leave Priority as Medium.
4. Tick "Create another" (bottom of the window) so the form stays open, then click Create.
5. Repeat for all 5 Epics, then click Cancel/close when the last one is created.
6. Press the E key on your keyboard (or click "Epic" above the backlog and switch on "Show epic panel"). You should now see 5 Epics on the left.

| Summary (copy exactly) | Description (copy exactly) | Priority | Type |
|---|---|---|---|
| Epic 1: Account Registration & Access | Enable trial candidates to register and all users to log in securely, with role-specific access for candidates, clinical researchers and administrators. | Medium | Epic |
| Epic 2: Clinical Trial Management | Enable clinical researchers to create, edit and close clinical trials and to define the eligibility criteria used to screen candidates. | Medium | Epic |
| Epic 3: Trial Discovery & Application | Enable trial candidates to find open clinical trials, read their details and submit applications that are automatically checked against the trial's eligibility criteria. | Medium | Epic |
| Epic 4: Application Review & Selection | Enable clinical researchers to review the applications submitted to their trials and record an Accepted or Rejected decision for each. | Medium | Epic |
| Epic 5: Security, Privacy & Performance | Protect candidate personal and health-related data and keep the system fast enough for normal use. | Medium | Epic |

## Step 3: Create the 17 User Stories (handout Step 3)

1. In the Epic panel on the left, click the small arrow next to Epic 1. Click "Create work item".
2. Check that Work type is Story. Paste the story's Summary from the tables below.
3. In Description, paste the three lines (As a... / I want... / So that...). Each line goes on its own line.
4. Scroll down to Priority and set it to the priority in the table (the default is Medium). Jira shows Highest, High, Medium, Low, Lowest. Only use High, Medium or Low.
5. Click Create. The story appears in the Backlog under that Epic. Repeat until all stories of that Epic are done, then do the next Epic.

**Note:** Creating a story from the Epic panel automatically links it to the correct Epic. If you create it with the main blue Create button, set the "Parent" field to the Epic yourself.

### Epic 1: Account Registration & Access

| Summary (copy) | Description (copy 3 lines) | Priority | Points | Sprint |
|---|---|---|---|---|
| Story 1.1: Candidate Registration | **As a** trial candidate,<br>**I want** to register an account with my email, a password and basic profile details,<br>**So that** I can apply for clinical trials. | High | 5 | Sprint 1 |
| Story 1.2: User Login | **As a** registered user (candidate, researcher or administrator),<br>**I want** to log in with my email and password,<br>**So that** I can access the features for my role. | High | 3 | Sprint 1 |
| Story 1.3: Account Lock-out After Failed Logins | **As a** trial candidate,<br>**I want** my account to be locked for 15 minutes after 5 failed login attempts,<br>**So that** other people cannot guess my password. | Medium | 2 | Backlog |
| Story 1.4: Manage User Accounts | **As a** system administrator (design assumption),<br>**I want** to create and deactivate researcher accounts,<br>**So that** only authorised researchers can create and manage trials. | Low | 3 | Backlog |

### Epic 2: Clinical Trial Management

| Summary (copy) | Description (copy 3 lines) | Priority | Points | Sprint |
|---|---|---|---|---|
| Story 2.1: Create Clinical Trial | **As a** clinical researcher,<br>**I want** to create a clinical trial with a title, description, location and closing date,<br>**So that** suitable candidates can discover and apply to it. | High | 5 | Sprint 1 |

| Summary (copy) | Description (copy 3 lines) | Priority | Points | Sprint |
|---|---|---|---|---|
| Story 2.2: Define Eligibility Criteria | **As a** clinical researcher,<br>**I want** to add eligibility criteria such as age range and required medical condition to a trial,<br>**So that** only suitable candidates are accepted for screening. | High | 5 | Sprint 1 |
| Story 2.3: Edit Clinical Trial | **As a** clinical researcher,<br>**I want** to edit the details of an existing trial,<br>**So that** candidates always see correct and up-to-date information. | Low | 2 | Backlog |
| Story 2.4: Close Clinical Trial | **As a** clinical researcher,<br>**I want** to close a trial when enough participants have been recruited,<br>**So that** no new applications are received for it. | Medium | 2 | Backlog |

## Epic 3: Trial Discovery & Application

| Summary (copy) | Description (copy 3 lines) | Priority | Points | Sprint |
|---|---|---|---|---|
| Story 3.1: View Open Trials and Details | **As a** trial candidate,<br>**I want** to view a list of open clinical trials and the details and eligibility criteria of each,<br>**So that** I can decide which trial to apply for. | High | 3 | Sprint 2 |
| Story 3.2: Submit Trial Application | **As a** trial candidate,<br>**I want** to submit an application for a selected trial with the required information,<br>**So that** the researcher can consider me for the trial. | High | 5 | Sprint 2 |
| Story 3.3: Automatic Eligibility Check | **As a** trial candidate,<br>**I want** the system to check my application against the trial's eligibility criteria automatically,<br>**So that** I know immediately whether I qualify and which criteria I do not meet. | High | 8 | Sprint 2 |
| Story 3.4: Application Form Validation | **As a** trial candidate,<br>**I want** missing or invalid fields to be highlighted before my application is saved,<br>**So that** I can correct mistakes and avoid submitting an incomplete application. | Medium | 3 | Backlog |

## Epic 4: Application Review & Selection

| Summary (copy) | Description (copy 3 lines) | Priority | Points | Sprint |
|---|---|---|---|---|
| Story 4.1: View Applications for My Trial | **As a** clinical researcher,<br>**I want** to view the applications submitted to my trials with each candidate's information and eligibility result,<br>**So that** I can review candidates efficiently. | High | 5 | Sprint 2 |
| Story 4.2: Record Application Decision | **As a** clinical researcher,<br>**I want** to mark each application as Accepted or Rejected,<br>**So that** the recruitment outcome is recorded for every candidate. | High | 3 | Backlog |

## Epic 5: Security, Privacy & Performance

| Summary (copy) | Description (copy 3 lines) | Priority | Points | Sprint |
|---|---|---|---|---|
| Story 5.1: Secure Data in Transit and Passwords | **As a** trial candidate,<br>**I want** my password and personal data to be protected with HTTPS and hashed password storage,<br>**So that** my sensitive information cannot be read by other people. | High | 3 | Sprint 1 |
| Story 5.2: Role-Based Access Control | **As a** trial candidate,<br>**I want** only authorised users to be able to see my health-related data,<br>**So that** my privacy is protected. | High | 5 | Backlog |

| Summary (copy) | Description (copy 3 lines) | Priority | Points | Sprint |
|---|---|---|---|---|
| Story 5.3: Fast Response Under Load | **As a** trial candidate,<br>**I want** trial pages and application submissions to respond within a few seconds,<br>**So that** I can complete my application without waiting. | Medium | 3 | Backlog |

Story points are set in Step 5. The Sprint column tells you which stories go into which sprint in Step 6.

## Step 4: Prioritise the backlog (handout Step 4)

1. Check every story shows the priority from the tables (High, Medium or Low). To change one, click the story, find "Priority" in the right-hand Details panel and pick the new value.
2. Reorder the backlog: drag stories up or down so they follow this order (rank 1 at the top of the Backlog list).

| Rank | Story | Priority | Points |
|---|---|---|---|
| 1 | Story 1.1: Candidate Registration | High | 5 |
| 2 | Story 1.2: User Login | High | 3 |
| 3 | Story 5.1: Secure Data in Transit and Passwords | High | 3 |
| 4 | Story 2.1: Create Clinical Trial | High | 5 |
| 5 | Story 2.2: Define Eligibility Criteria | High | 5 |
| 6 | Story 3.1: View Open Trials and Details | High | 3 |
| 7 | Story 3.2: Submit Trial Application | High | 5 |
| 8 | Story 3.3: Automatic Eligibility Check | High | 8 |
| 9 | Story 4.1: View Applications for My Trial | High | 5 |
| 10 | Story 4.2: Record Application Decision | High | 3 |
| 11 | Story 5.2: Role-Based Access Control | High | 5 |
| 12 | Story 3.4: Application Form Validation | Medium | 3 |
| 13 | Story 1.3: Account Lock-out After Failed Logins | Medium | 2 |
| 14 | Story 2.4: Close Clinical Trial | Medium | 2 |
| 15 | Story 5.3: Fast Response Under Load | Medium | 3 |
| 16 | Story 2.3: Edit Clinical Trial | Low | 2 |
| 17 | Story 1.4: Manage User Accounts | Low | 3 |

## Step 5: Assign story points (handout Step 5)

Real teams play "planning poker" together. You are working alone, so think of it as asking yourself: how big is this compared with the others? The report already explains each estimate.

1. Click a story in the Backlog. A panel opens on the right.
2. Scroll down in the panel to "More fields" and open it. Find "Story Points".
3. Type the number from the tables (for example 5), then click the tick mark to save.
4. Repeat for all 17 stories. Each story row in the Backlog now shows its number.
5. Click the arrow next to an Epic in the Epic panel. It now shows the total points of that Epic. Check the totals: Epic 1 = 13, Epic 2 = 14, Epic 3 = 19, Epic 4 = 8, Epic 5 = 11 (total 65).

**Screenshot 1:** Backlog with the Epic panel open and all Epics and stories visible (priority icons visible).
**Screenshot 2:** Backlog showing the story points on each story, plus one Epic expanded in the Epic panel showing its total points.

## Step 6: Run Sprint 1 (handout Step 6)

**Sprint 1 goal: Users can register and log in securely, and researchers can create trials with eligibility criteria.**

1. On the Backlog tab, find the box at the top called "CTRE Sprint 1".
2. Tick the check boxes of these 5 stories (21 points in total), then drag them into the Sprint 1 box:
- **Story 1.1: Candidate Registration** (5 points)
- **Story 1.2: User Login** (3 points)
- **Story 5.1: Secure Data in Transit and Passwords** (3 points)
- **Story 2.1: Create Clinical Trial** (5 points)
- **Story 2.2: Define Eligibility Criteria** (5 points)
3. Click "Start sprint". Sprint name: CTRE Sprint 1. Duration: 1 week. Start date: now. Sprint goal: paste the goal above. Click Start.
4. Jira opens "Active sprints" with all 5 stories in the To Do column.
5. Simulate the work by dragging cards between columns (To Do → In Progress → Done) following the schedule below. If you do everything in one sitting, wait a few minutes between moves so the burndown chart shows separate steps.

| Day | What to move | Points left |
|---|---|---|
| Day 1 | Start sprint. Move 1.1 and 1.2: To Do → In Progress. | 21 |
| Day 2 | 1.1 and 1.2: In Progress → Done. Move 5.1: To Do → In Progress. | 13 |
| Day 3 | 5.1: In Progress → Done. Move 2.1: To Do → In Progress. | 10 |
| Day 4 | 2.1: In Progress → Done. Move 2.2: To Do → In Progress. | 5 |
| Day 5 | 2.2: In Progress → Done. Complete sprint. | 0 |

**Screenshot 3:** Active sprints board of Sprint 1 while cards are in different columns, and again (or instead) with all cards in Done.
**Screenshot 4 (take it BEFORE you click Complete sprint):** the Sprint 1 burndown chart (see Step 8).

6. When all 5 cards are in Done, click "Complete sprint" (top right of the board), then click "Complete sprint" again in the pop-up. Jira takes you back to the Backlog.

## Step 7: Run Sprint 2

**Sprint 2 goal: Candidates can view open trials and apply with an automatic eligibility check, and researchers can view applications.**

1. In the Backlog, use the "CTRE Sprint 2" box at the top. If there is none, click "Create sprint".
2. Tick and drag these 4 stories into it (21 points in total):
- **Story 3.1: View Open Trials and Details** (3 points)
- **Story 3.2: Submit Trial Application** (5 points)
- **Story 3.3: Automatic Eligibility Check** (8 points)
- **Story 4.1: View Applications for My Trial** (5 points)
3. Click "Start sprint". Sprint name: CTRE Sprint 2. Duration: 1 week. Paste the Sprint 2 goal. Click Start.
4. Move the cards on the Active sprints board following this schedule:

| Day | What to move | Points left |
|---|---|---|
| Day 1 | Start sprint. 3.1: To Do → In Progress → Done. Move 3.2: To Do → In Progress. | 18 |
| Day 2 | 3.2 stays In Progress. | 18 |
| Day 3 | 3.2: In Progress → Done. Move 3.3: To Do → In Progress. | 13 |
| Day 4 | 3.3 stays In Progress. Move 4.1: To Do → In Progress. | 13 |
| Day 5 | 3.3 and 4.1: In Progress → Done. Complete sprint. | 0 |

**Screenshot 5:** Active sprints board of Sprint 2 (cards in different columns, or all in Done).
**Screenshot 6 (before clicking Complete sprint):** the Sprint 2 burndown chart.

5. Click "Complete sprint" when all 4 cards are Done. The 8 stories you did not select stay in the Backlog for a later sprint (23 points).

## Step 8: Generate the Burndown Chart (handout Step 7)

1. In the top menu, click "Reports" (next to Calendar). If you do not see it, click "More".
2. Choose "Burndown Chart".
3. At the top left of the chart, use the sprint drop-down to pick "CTRE Sprint 1". Make sure the second drop-down says "Story Points".
4. Take the screenshot. Then pick "CTRE Sprint 2" and take the second screenshot.
5. Read the chart: the grey line (Guideline) is where you should be if work is finished evenly. The red line (Remaining values) is what is really left. Red below grey = ahead of plan.

**Note:** Jira draws the chart from the real time you moved the cards, so it will not look exactly like the simulated charts in Appendix A of the report. That is fine: describe what YOUR chart shows. If your real chart is very different (for example a single big drop because you moved all cards at once), edit the Q3 and Q4 answers in the report so they match what you see.

## Step 9: Finish the report

1. Open SE_Lab2_Jira_Report.docx in Microsoft Word.
2. Go to Section 7. In each dashed box, delete the grey text and paste (Ctrl+V) or insert the matching screenshot: Figure 1 = Screenshot 1, Figure 2 = Screenshot 2, Figure 3 = Screenshot 3, Figure 4 = Screenshot 5, Figure 5 = Screenshot 4, Figure 6 = Screenshot 6. Resize each picture to the page width.
3. Read Section 8 (Reflection). Check each answer matches what really happened in your Jira. Edit the sentences that do not.
4. Check your SRN (PES1UG25CS845) appears in the header, then use File → Save As → PDF (or Export → Create PDF) to make the final PDF.

## Final checklist and instructor demo

The handout says you must show your Jira workspace to the instructor. Before the demo, make sure you can open each of these quickly:

- Backlog with 5 Epics and 17 stories, priorities and story points visible
- Epic panel with the points totals
- Active sprints board (Sprint 2 is the most recent) and the completed Sprint 1
- Reports → Burndown Chart for both sprints
- Report PDF with 6 screenshots and the reflection answers

## Troubleshooting

- **Story Points field is missing:** open the story, scroll the right panel to "More fields". If it is still missing, ask your instructor or look under Project settings → Fields.
- **Jira warns that some items have no estimate when starting a sprint:** you skipped a story's points. Cancel, add the points, and start again.
- **No Sprint 2 box in the Backlog:** click "Create sprint" in the Backlog section.
- **The burndown chart is empty or flat:** the sprint needs to be started and have estimated stories. Check the sprint drop-down and that the estimate is Story Points.
- **A story ended up in the wrong Epic:** click the story, find "Parent" in the Details panel and choose the correct Epic.